Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2024-I-Q03] | Part A (Q1-Q5)

Two players alternately remove either 1 or 4 tokens from a pile, with the last removal winning. The first player moves first. For how many initial sizes $1 \le n \le 2024$ can the second player force a win?

两人轮流从一堆筹码中取走 1 或 4 枚，取走最后一枚者获胜。求 $1 \le n \le 2024$ 中有多少个初始筹码数能保证后手获胜。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2010-I-Q12] | Part C (Q11-Q15)

Find the least $m \geq 3$ such that every two-coloring of $\{3, 4, \ldots, m\}$ contains same-colored numbers a, b, c, not necessarily distinct, with $ab = c$.

求最小整数 $m \geq 3$，使 $\{3, 4, \ldots, m\}$ 的任意二染色中，必有同色的 a, b, c 满足 $ab = c$，三数允许重复。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 6.1 Games, invariants & state strategies

博弈、不变量与状态策略

[AIME-2012-I-Q11] | Part C (Q11-Q15)

From $(0,0)$, a frog may repeatedly add $(7,2),(2,7),(-5,-10),(-10,-5)$. Let M count the reachable integer points with $|x|+|y|\le 100$. Find $M \bmod 1000$.

青蛙从原点出发，每跳可加向量 $(7,2),(2,7),(-5,-10),(-10,-5)$ 之一。设可达且满足 $|x|+|y|\le 100$ 的整数点有 M 个，求 M 模 1000。

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